

Practical – 7

Task –1 Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
GNU nano 8.7.1                                memory_layout.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    int stack_var = 30;
    static int local_static_var = 40;

    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n\n", getpid());

    printf("Code address      : %p\n", (void *)code_function);
    printf("Global variable   : %p\n", (void *)&global_var);
    printf("Static variable    : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static_var);
    printf("Heap variable      : %p\n", (void *)heap_var);
    printf("Stack variable     : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press Ctrl+C to stop.\n");

    while (1)
    {
        sleep(2);
    }

    free(heap_var);

    return 0;
}
```

```

gopinath@GOPINATH:~$ mkdir practical_memory
gopinath@GOPINATH:~$ cd practical_memory
gopinath@GOPINATH:~/practical_memory$ nano memory_layout.c
gopinath@GOPINATH:~/practical_memory$ gcc memory_layout.c -o memory_layout
gopinath@GOPINATH:~/practical_memory$ ./memory_layout
Process ID: 757

Code address      : 0x55c67c7d31c9
Global variable   : 0x55c67c7d6010
Static variable   : 0x55c67c7d6014
Local static variable : 0x55c67c7d6018
Heap variable     : 0x55c6a55f9010
Stack variable    : 0x7ffc49ae650c

Process is running...
Press Ctrl+C to stop.
^C
gopinath@GOPINATH:~/practical_memory$

```

Task-2 Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void display()
{
    printf("Inside code segment\n");
}

int main()
{
    int stack_var = 30;
    static int local_static = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n", getpid());

    printf("Code address      : %p\n", (void *)display);
    printf("Global variable   : %p\n", (void *)&global_var);
    printf("Static variable    : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static);
    printf("Heap variable      : %p\n", (void *)heap_var);
    printf("Stack variable     : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");

    while (1)
    {
        sleep(2);
    }

    free(heap_var);

    return 0;
}

```

```

gopinath@GOPINATH:~$ nano memory_analysis.c
gopinath@GOPINATH:~$ nano memory_analysis.c
gopinath@GOPINATH:~$ ^C
gopinath@GOPINATH:~$ gcc memory_analysis.c -o memory_analysis
gopinath@GOPINATH:~$ ./memory_analysis &
[1] 1017
Process ID: 1017
Code address      : 0x5d9c9ddb91c9
Global variable   : 0x5d9c9ddb7010
Static variable   : 0x5d9c9ddb7014
Local static variable : 0x5d9c9ddb7018
Heap variable     : 0x5d9cadd8f010
Stack variable    : 0x7ffd639b5ebc

Process is running...
gopinath@GOPINATH:~$ cat /proc/1017/maps
5d9c9ddb3000-5d9c9ddb4000 r--p 00000000 08:30 2739 /home/gopinath_priya/memory_analysis
5d9c9ddb4000-5d9c9ddb5000 r-xp 00001000 08:30 2739 /home/gopinath_priya/memory_analysis
5d9c9ddb5000-5d9c9ddb6000 r--p 00002000 08:30 2739 /home/gopinath_priya/memory_analysis
5d9c9ddb6000-5d9c9ddb7000 r--p 00003000 08:30 2739 /home/gopinath_priya/memory_analysis
5d9c9ddb7000-5d9c9ddb8000 rw-p 00003000 08:30 2739 /home/gopinath_priya/memory_analysis
5d9cad3f0000-5d9cad3f3000 rw-p 00000000 00:00 0 [heap]
7a5d0c600000-7a5d0c628000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c628000-7a5d0c7c0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c7c0000-7a5d0c8e0000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c8e0000-7a5d0c812000 r--p 002d0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c812000-7a5d0c814000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c814000-7a5d0c821000 rw-p 00000000 00:00 0
7a5d0c892000-7a5d0c893000 rw-p 00000000 00:00 0
7a5d0c892000-7a5d0c89e000 rw-p 00000000 00:00 0
7a5d0c899000-7a5d0c89a000 r--p 00000000 00:00 0 [vvar]
7a5d0c89a000-7a5d0c8a0000 r-xp 00000000 00:00 0 [vvar_vclock]
7a5d0c8a0000-7a5d0c8c1000 r--p 00000000 08:30 13625 [vdso]
7a5d0c8c1000-7a5d0c8c7000 r-xp 00021000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c8c7000-7a5d0c8cc000 r--p 00077000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c8cc000-7a5d0c8ce000 r--p 0007b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c8ce000-7a5d0c8d0000 rw-p 0007d000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffd63970000-7ffd63998000 rw-p 00000000 00:00 0
7ffd63998000-7ffd6399b000 r-xp 00000000 00:00 0
gopinath@GOPINATH:~$ █

```